



[The exam text is in English. You should answer the questions in English.]

The exam consists of 13 questions (6 pages). Please first check if any page is missing in the printout you. For the open questions: please keep the handwriting clear and readable.

The exam starts at 9.00 hrs and ends at 12.00 hrs. Participation in the exam requires being present for **at least 1 hour**.

Electronic devices are not allowed and must be stowed.
Points for each open question are indicated.

- Please write readable and write your name and student identification number on every single sheet of paper for unique identification!
- **Don't spend much time writing long stories – that's NOT necessary. Don't panic! You CAN make it!**
- Look at the whole exam first to get an overview, pay attention to the grading scheme of multiple-choice questions, and do those assignments first which you feel are doable for you.
- Good success and all the best for your studies!

Part I - Multiple-choice [50 points].

- Each problem has a variable number of correct statements. Please write your answers in the table given at the end of the question sheet.
- In NO problem, all statements are correct. If you choose all the options, i.e., "ABCD", the score for that question will be ZERO!
- The score for each question will be given according to Morgan method:

$$5 \times \max \{ r / \#correct\ option(s) - w / \#distractors(s), 0 \},$$

where:

- 5 is the total score for each question;
- r is the number of option(s) you select correctly;
- w is the number of option(s) you select incorrectly;
- $\#correct\ option(s)$ is the number of correct options of this question;
- $\#distractors(s)$ is the number of incorrect options/distractors of this question;
- Please DO NOT mark an option if you are not certain with it as a penalty will incur.

Please write the alphabetic characters (e.g., CD) representing your answers of the multiple-choice questions into the table below.

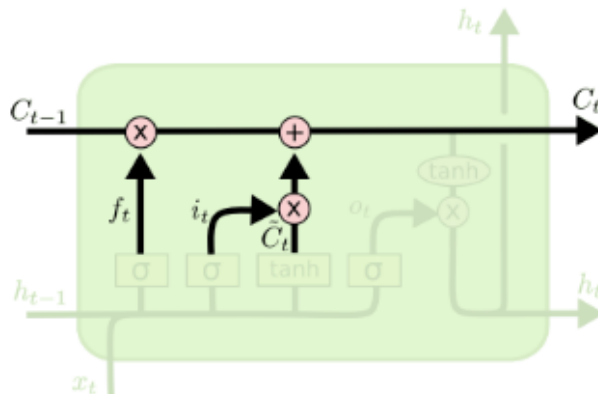
No.	1	2	3	4	5	6	7	8	9	10
Answers										

1. Regarding the biological and McCulloch-Pitts models, which of the following descriptions are correct? [more than one answer is correct]
 - a. Synaptic strength of the biological neuron is translated into the bias of the McCulloch-Pitt model.
 - b. A step function/threshold unit is used as the activation function in the McCulloch-Pitt model.
 - c. Excitation refers to the negative product of the incoming input and the corresponding synaptic strength.
 - d. Spikes of a biological neuron can be interpreted as firing rates.
2. What is the wrong statement about a Hopfield network? [one answer is correct]
 - a. Hopfield network contains loops in its graphic representation.
 - b. Attractors are local minima of the energy function.
 - c. Hopfield network has multiple layers of neurons.
 - d. Those two models should be trained using different algorithms.
3. Which of the following statements is true about Hebbian learning? [one answer is correct]
 - a. Hebb's rule says that the weight between two neurons should be decreased if they are activated simultaneously.
 - b. The learning rate in Hebbian learning is usually larger than one.
 - c. We need apply Hebb's rule on each data point and pass over the entire training set.
 - d. The Hebbian learning process will always stabilize after passing the training set several times.
4. In binary classification tasks using a simple perceptron, which of the following statements are true? [more than one answer is correct]
 - a. The margin is defined as the distance of the closest point to the decision boundary among all correctly classified ones.
 - b. The margin is defined as the distance of the furthest point to the decision boundary among all correctly classified ones.
 - c. Finding the optimal weight vector is translated into a maxmin problem.
 - d. It is often convenient to ask that the sign of the predicted and the expected output is smaller than a margin.
5. Which of the following statements are true about multi-layer perceptron? [more than one answer is correct]
 - a. It is well-suited for image classification tasks.
 - b. The XOR problem cannot be solved by multi-layer perceptron.
 - c. Multi-layer perceptron is typically trained by performing gradient descent method.
 - d. The functionality of the hidden layer is to extract features.
6. Which of the following statements are true regarding the backpropagation (BP) algorithm and batch method? [more than one answer is correct]
 - a. The BP algorithm consists of a forward phase and a backward phase.
 - b. Consider a simple perceptron $O = h(w \cdot x + b)$. We can expand the partial derivative $\partial O / \partial w$ as $\partial O / \partial h \cdot \partial h / \partial w$.
 - c. Batch size is the number of data points taken to compute the loss value in each iteration of the BP algorithm.
 - d. An epoch is when half the dataset is passed through the neural network.

7. Which of the following tasks would be more suitable to use RNNs? [more than one answer is correct]
- Forecast the traffic of the next day at a road crossing, given 30-day history thereof.
 - Predict the next word of a sentence based on the previous words.
 - Detect real-world objects in images.
 - Detect anomalous events in bank transactions.
8. Which of the following statements are true about CNNs? [more than one answer is correct]
- In a handwritten digits recognition task (i.e., MNIST), we use the sigmoid unit to obtain the classification probability.
 - Kernels in CNN models are also weights which will be learned via back-propagation process.
 - We can use the cross entropy loss function if the output of the CNN model is classification probability.
 - A tensor is a multi-dimensional array.
9. Which statements are correct about convolution and pooling? [more than one answer is correct]
- Convolution and pooling both contain trainable parameters.
 - Convolution and pooling both extract features from the input image.
 - Convolution and pooling both increases the dimension/size of the input image.
 - Multiple convolutions are performed in each layer of a CNN in order to capture different features of the input.
10. Please calculate the size of the output image (assuming the stride is one) for a given input image and kernel size. Case (1): input size 7x7, kernel size 3x3; case (2): input size 18x18, kernel size 4x4. [more than one answer is correct]
- For case 1, if no padding, the size of the output matrix is 5x5.
 - For case 1, if the padding size is (2, 2), the size of the output volume is 6x6.
 - For case 2, if the padding size is (3, 3), the size of the output volume is 20x20.
 - For case 2, if no padding, the size of the output volume is 15x15.

Part II - Open Questions [50 points]

1. (15 points) Consider the following diagram representing a Long Short-Term Memory (LSTM) unit.



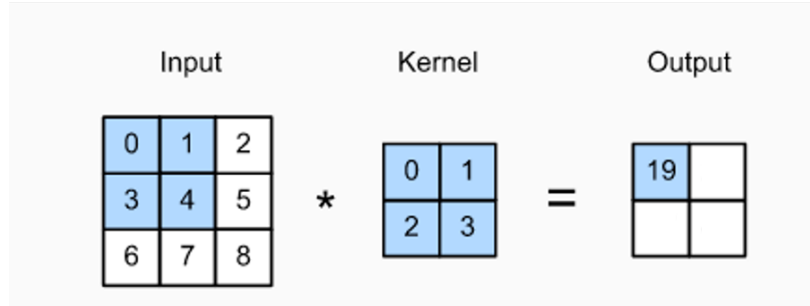
- a) (5) What is the functionality of the variable C_t in the above diagram? Please write it down with your own words.
- b) (5) Please write down the expression or operation of the forget gate with the symbols in the above diagram.
- c) (5) In contrast to the vanilla Recurrent neural network (RNN), the LSTM relaxes the well-known issue of vanishing/exploding gradients. Please describe in your own words, how LSTM manages to achieve it.



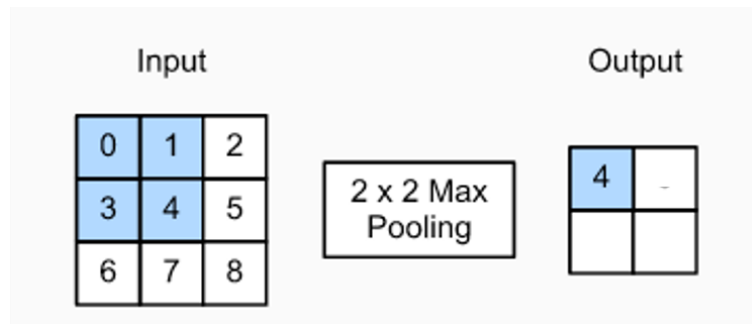
2. **(20 points)** Consider a Hopfield network, whose main task is to address the associative memory problem (store a set of p patterns $\vec{\xi}^1, \dots, \vec{\xi}^p$ such that when presented with a new pattern $\vec{x}$, the network responds by producing whichever one of the stored patterns that most closely resembles $\vec{x}$) and perform the following tasks.
- (5)** Give a graphical representation of a stored pattern $\vec{\xi}^i = [-1, -1, 1, -1, 1]$.
 - (5)** Considering the case of one stored pattern $\vec{\xi}$, what is the condition for the pattern to be a stable pattern, and how should the weights be defined to guarantee that?
 - (5)** Prove that the definition of the weights given in (b) makes the pattern $\vec{\xi}$ stable.
 - (5)** Design a network of 3 neurons (i.e., compute the weights) that remembers a pattern $\vec{\xi} = [1, -1, 1]$. Then prove that $\vec{\xi}$ is stable for the designed network.

3. **(15 points)** Consider the convolution and pooling operations and answer the following questions:

a. **(5)** Please complete the output of the convolution given in the figure below.



b. **(5)** Please complete the output of the 2 x 2 max-pooling in the figure below.



c. **(5)** Please specify [either graphically or by text description] how to apply the convolution operation to a multi-channel input image, e.g., the red, green, and blue channels in RGB images.